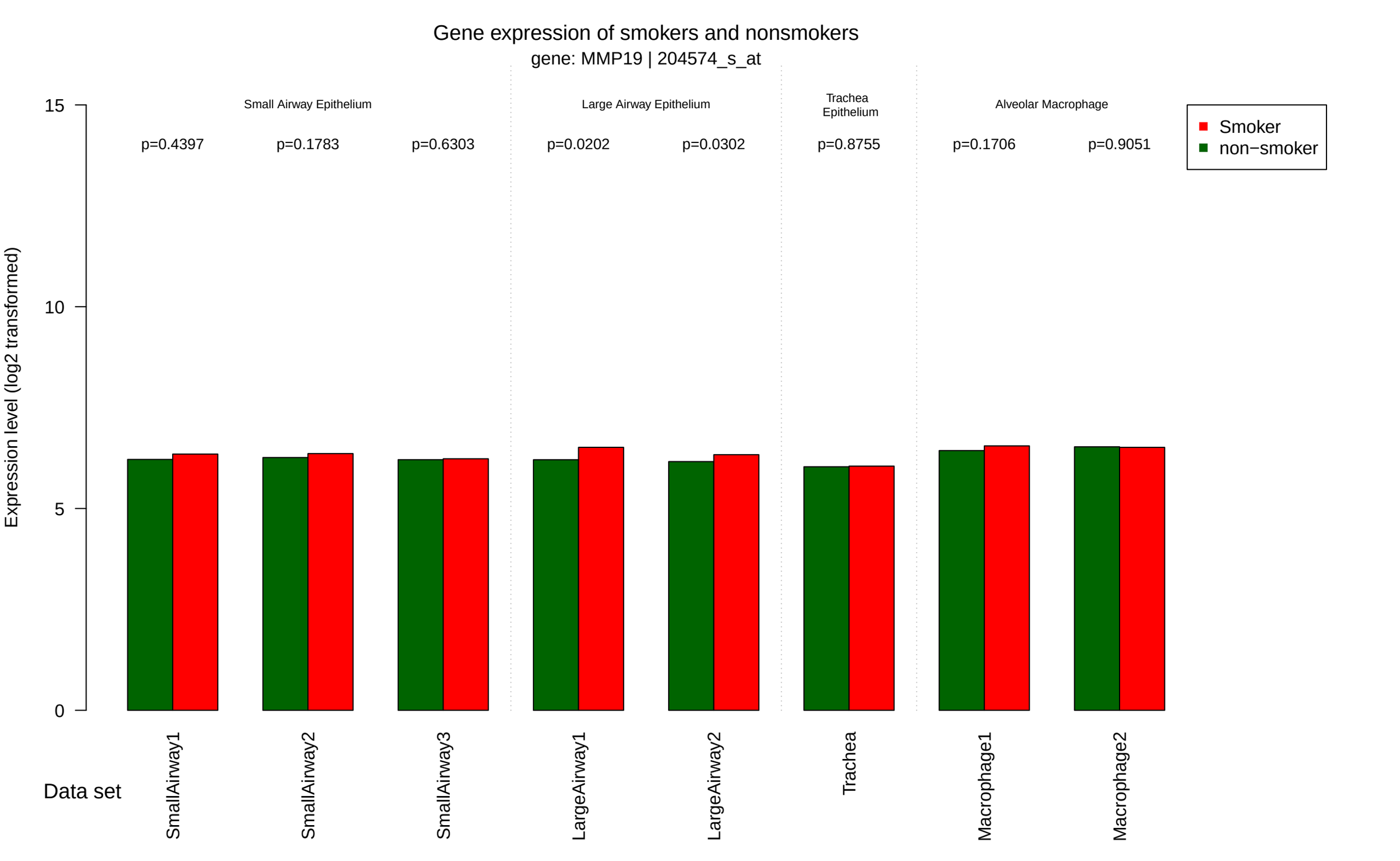
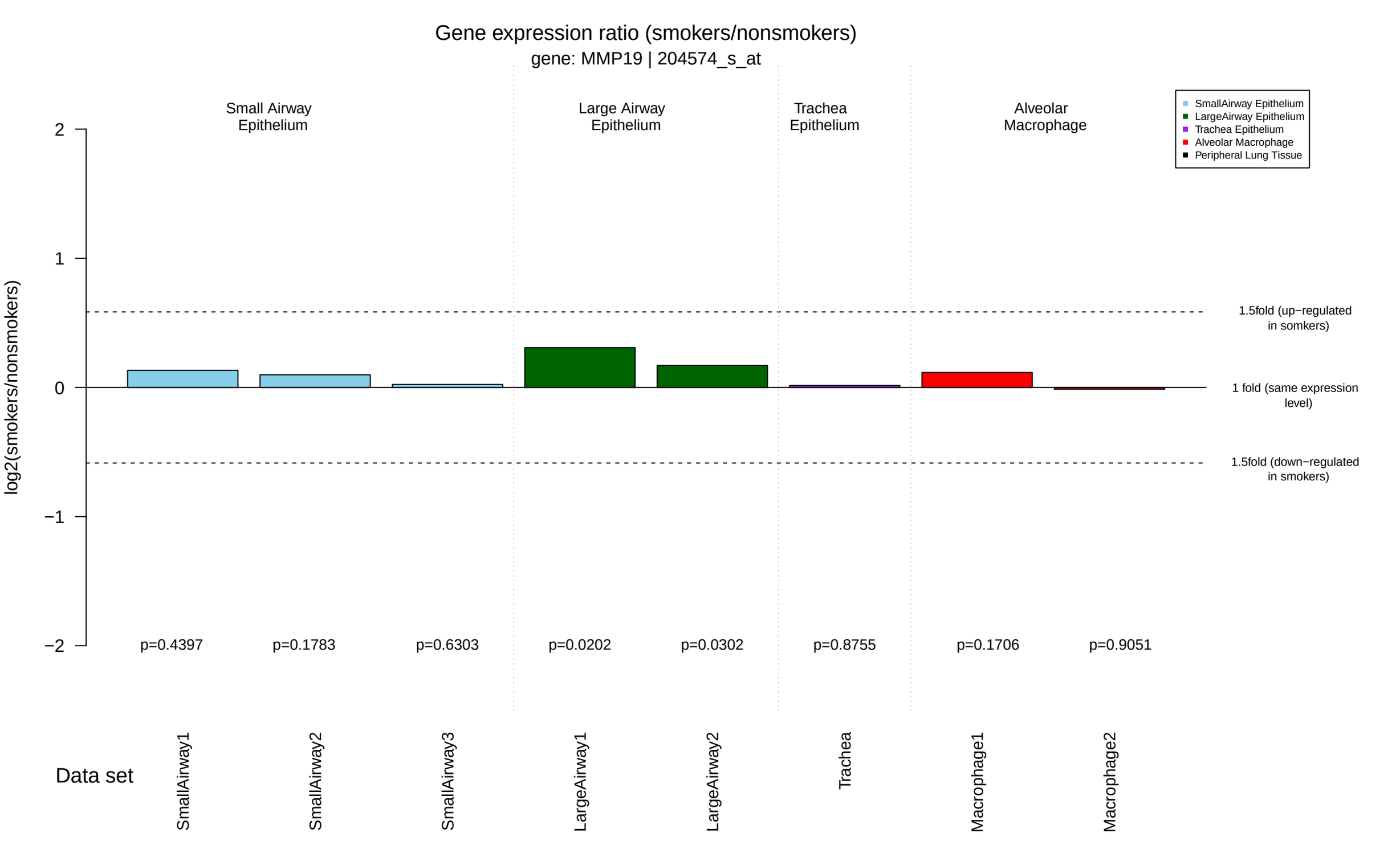
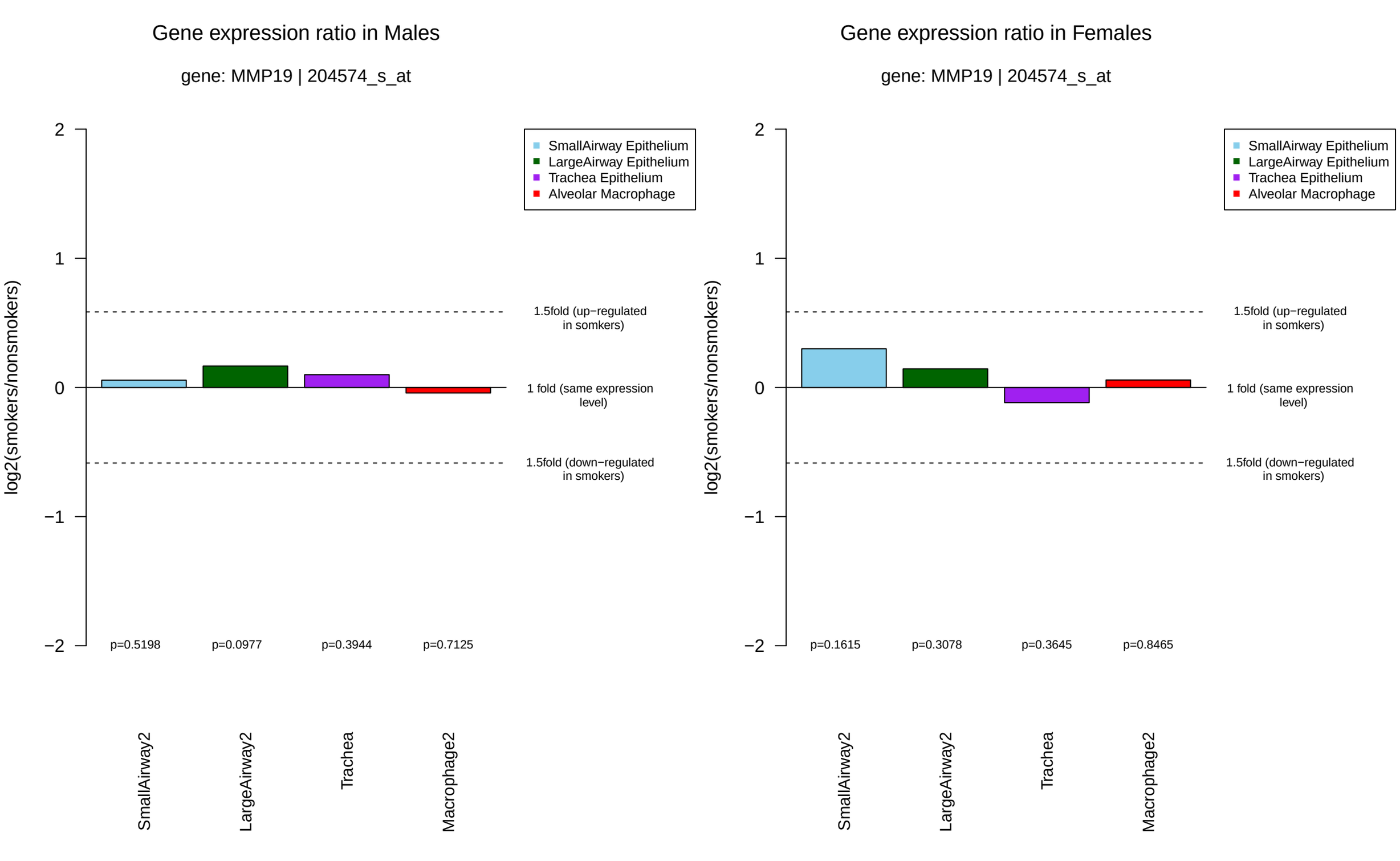
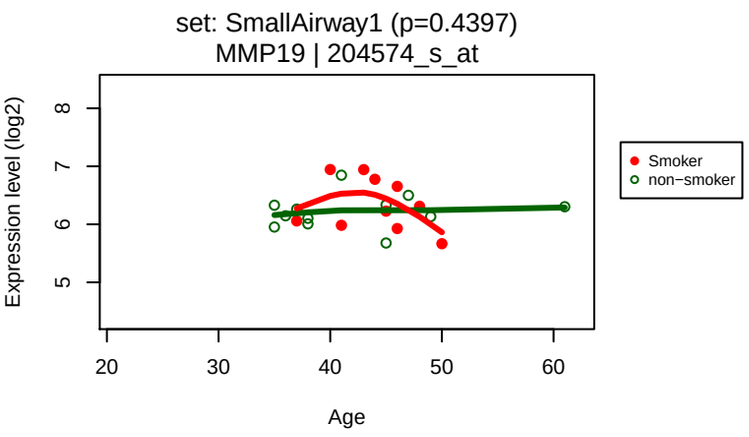
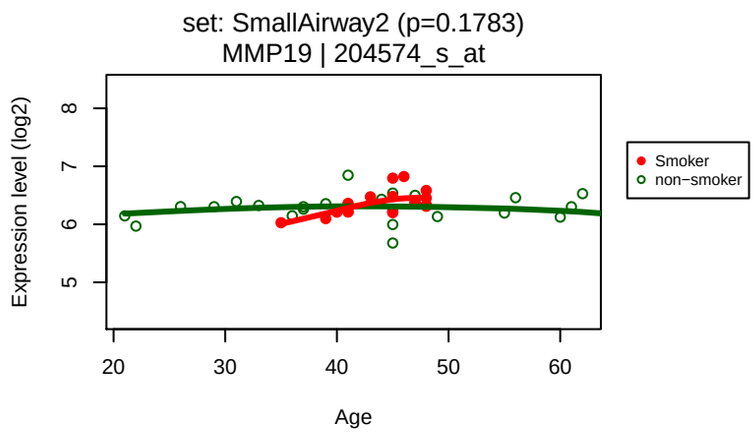
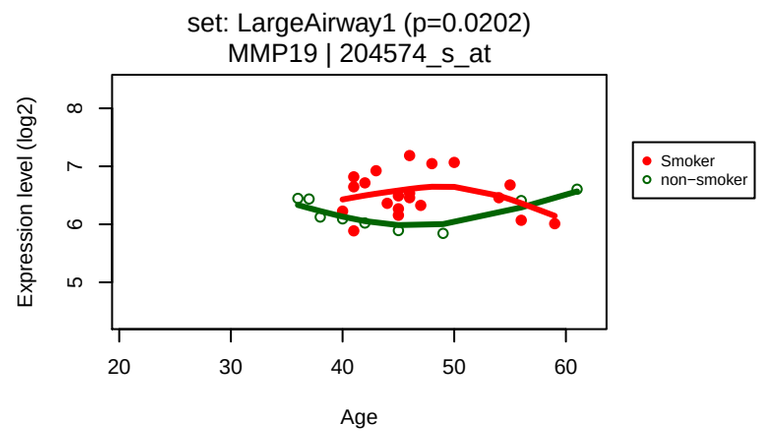

Gene expression of smokers and nonsmokers
gene: MMP19 | 204574_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4397
p=0.1783
p=0.6303
p=0.0202
p=0.0302
p=0.8755
p=0.1706
p=0.9051
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

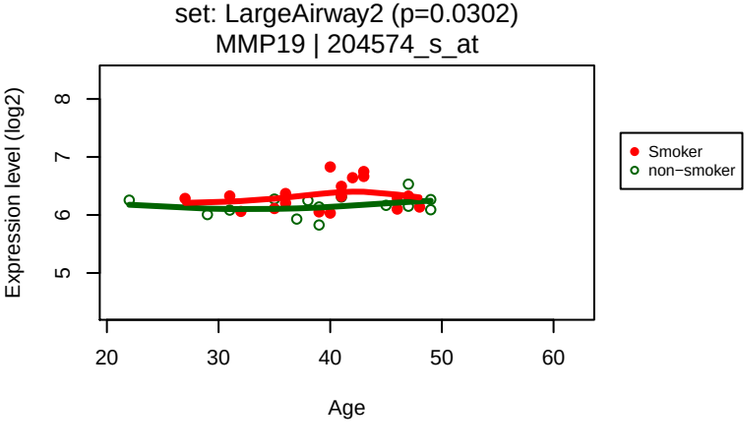

Gene expression ratio (smokers/nonsmokers)
gene: MMP19 | 204574_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4397
p=0.1783
p=0.6303
p=0.0202
p=0.0302
p=0.8755
p=0.1706
p=0.9051
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

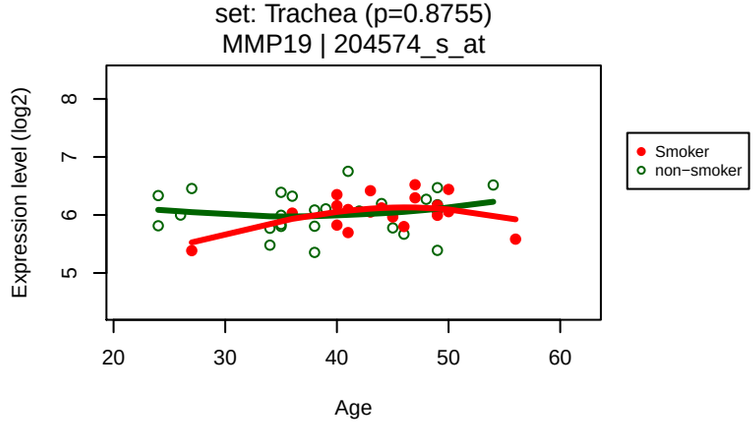

Gene expression ratio in Males
gene: MMP19 | 204574_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5198
p=0.0977
p=0.3944
p=0.7125
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP19 | 204574_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1615
p=0.3078
p=0.3645
p=0.8465
SmallAirway2
LargeAirway2
Trachea
Macrophage2

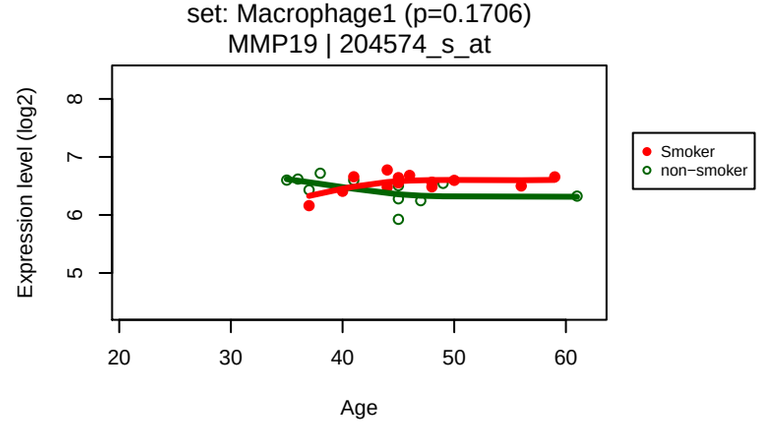

set: SmallAirway1 (p=0.4397)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker

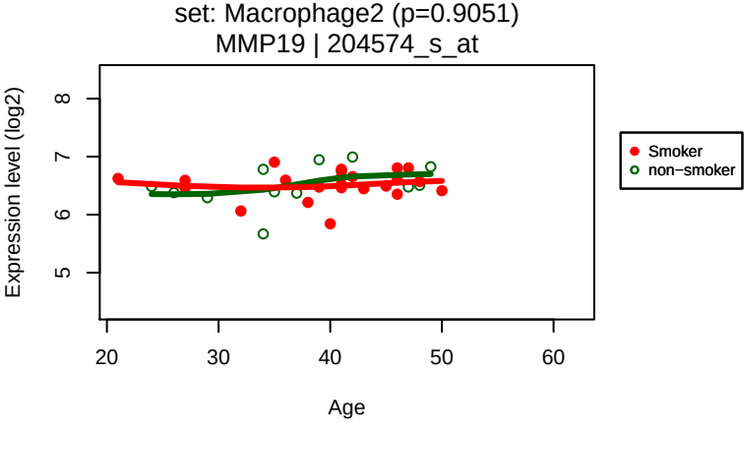

set: SmallAirway2 (p=0.1783)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker

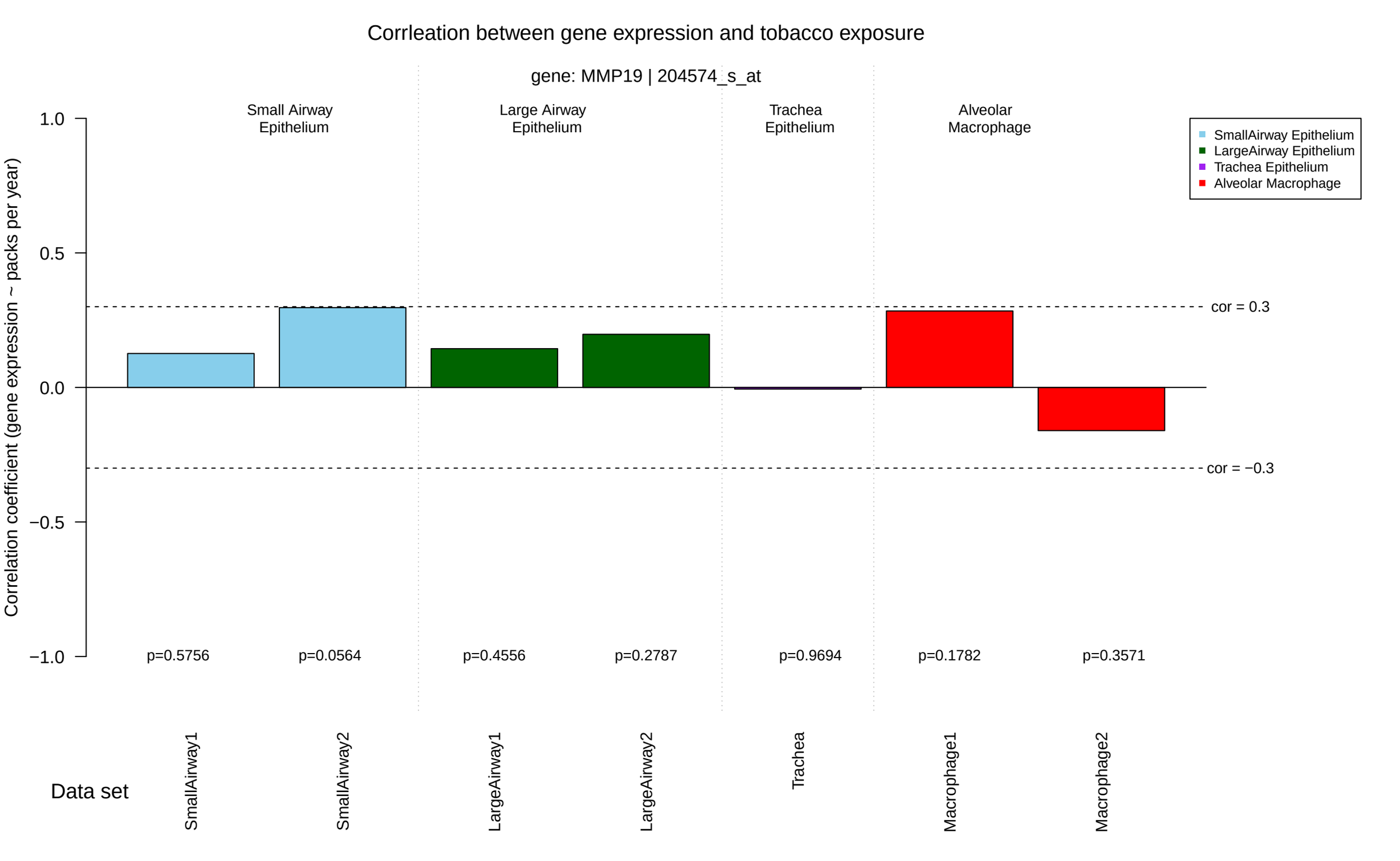

set: LargeAirway1 (p=0.0202)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0302)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8755)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1706)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9051)
MMP19 | 204574_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP19 | 204574_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5756
p=0.0564
p=0.4556
p=0.2787
p=0.9694
p=0.1782
p=0.3571
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2